

12-Head MHSA Accelerator — FPGA Hardware Validation Report

ViT-B/16 Multi-Head Self-Attention IP on PYNQ-ZU (XCZU7EV) — Progress Update

1. Objective

12-head MHSA IP (ViT-B/16 config), deploy it on FPGA, and validate hardware correctness against the PyTorch golden reference.

2. Configuration

Parameter	Value
Platform	PYNQ-ZU (Zynq UltraScale+ XCZU7EV)
Flow	Vitis HLS 2025.2 + Vivado IP Integrator
Heads	12
Sequence length	197
Head dim / Embed dim	64 / 768
Target clock	10 ns (100 MHz)

3. Steps Completed

- HLS: implemented 12-head MHSA kernel in C++, defined AXI-M ports (Q,K,V,O) and AXI-Lite control + scale register.
- Ran C-Simulation and RTL Co-Simulation; HLS synthesis completed successfully.
- Exported RTL as Vivado IP; built Zynq block design (PS + mhsa_top_0 + 4x AXI SmartConnect + reset block).
- Generated bitstream, exported hardware, generated and loaded PYNQ overlay on board — verified clean load.
- Read register map from PYNQ; verified control, Q/K/V/O pointer and scale registers via direct MMIO read/write (Untitled7.py).
- Programmed 64-bit physical addresses of allocated Q/K/V/O buffers into AXI-Lite registers; issued AP_START; polled AP_DONE.
- Bring-up tests: (a) constant-value inputs, (b) sparse single-nonzero input, (c) repeated runs — accelerator completed and output buffer changed in all cases, confirming AXI-M read/write is functional.
- Measured hardware latency over 10 repeated runs (Untitled7.py, cell 59).
- Ran post-implementation analysis in Vivado: timing, power, resource, DSP/BRAM/URAM usage.
- Began numerical verification: loaded q.txt/k.txt/v.txt and golden_output.txt, compared HW output vs PyTorch golden — mismatch currently under investigation (see Section 8).

4. Resource Utilization

HLS estimate (C-synthesis) vs. post-implementation (place & route):

Resource	HLS Estimate	Used (Place & Route)	Available	Utilization %
LUT	50,462	22,453 (25,191 CLB LUT)	230,400	~10.9%
FF	22,936	14,768 (18,590 CLB Reg)	460,800	~4.0%
DSP	527	525	1,728	30.4%
BRAM (18K)	6	35 (17.5 tiles)	312 tiles	5.6%
URAM	32	32	96	33.3%

5. Timing Summary

Metric	Setup	Hold	Pulse Width
Worst Slack	0.793 ns	0.010 ns	3.656 ns
Total Negative Slack	0.000 ns	0.000 ns	0.000 ns
Failing Endpoints	0	0	0
Total Endpoints	75,528	75,528	22,543

6. Power Summary

Metric	Value
Total on-chip power	3.606 W
Dynamic power	2.907 W (81%)
Static power	0.699 W (19%)
PS (processing system)	2.217 W (75% of dynamic)
PL logic + DSP + BRAM/URAM	0.690 W (~25% of dynamic)
Junction temperature	28.5 °C
Confidence level	Medium (vectorless estimate)

7. Measured Hardware Latency

Metric	Value
Average latency	14.305 ms
Minimum latency	14.294 ms
Maximum latency	14.346 ms

8. Throughput

Computed as $FLOPs \div \text{latency}$, using the same convention as the single-head baseline ($FLOPs = 4 \cdot N^2 \cdot D$, i.e. $2 \text{ matmuls} \times 2 \text{ flops/MAC}$, for QK^T and $Attn \cdot V$ across all 12 heads combined, $D = 768$):

Metric	Value
Total compute per inference	119.22 MFLOPs ($4 \cdot 197^2 \cdot 768$)
Throughput @ avg latency (14.305 ms)	8.33 GOPS
Throughput @ min latency (14.294 ms)	8.34 GOPS
Throughput @ max latency (14.346 ms)	8.31 GOPS
Tokens/s	~13,770
Elements/s	~10.58 Million
Throughput @ HLS-estimated latency (16.2–16.5 ms)	7.23 – 7.35 GOPS

9. Pending Work / Open Issue

- Numerical correctness (hardware output vs. PyTorch golden reference) is not yet verified for the 12-head design — a mismatch is currently being investigated.
- Root cause not yet isolated; candidates to check next: fixed-point scaling/quantization at head boundaries, per-head output buffer addressing/striding, and scale-factor register value (currently fixed at 512).
- Once corrected, re-run full error metrics (max/mean/RMSE) across all 12 heads and all 197 tokens, matching the methodology used for the single-head baseline.

10. Comparison with Single-Head Baseline

Metric	12-Head MHSA (Current)
Heads	12
Input / Output precision	fixed-point (int16 in / int32 accumulate, HW test)
LUT utilization	22,453 (~10.9% CLB LUT)
DSP utilization	525 (30.4%)
URAM utilization	32 (33.3%)
Timing (WNS)	0.793 ns met
Total power	3.606 W
Average latency	14,305 μ s (14.305 ms)
Throughput	8.33 GOPS
Functional correctness	Pending — mismatch under debug

11. Conclusion

The 12-head MHSA accelerator has been fully deployed end-to-end on FPGA: HLS design, IP packaging, Vivado integration, bitstream generation, overlay loading, register/AXI validation, and hardware execution are all working, with timing closed and resource/power figures characterized. Compared to the single-head baseline, resource and power usage scale as expected with 12 heads. The remaining task before this can be reported as a complete result is closing the gap on numerical correctness against the golden reference, which is the current focus.